PAYLOAD-AWARE TWO-TIER UAV ROUTE PLANNING FOR LAST-MILE DELIVERY

PHASE I - REVIEW I REPORT

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BONAFIDE CERTIFICATE

Certified that this project report titled **"PAYLOAD-AWARE TWO-TIER UAV ROUTE PLANNING FOR LAST-MILE DELIVERY"** is the *bonafide* work of "**THIRUMURUGAN RA (3122235001149)**, **SHRIKA SP THOTA (3122235001125)**, and **MEGHNA MANIMARAN (3122235001079)**" who carried out the project work under my supervision.

Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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ABSTRACT

UAV systems for medical and emergency last-mile delivery must balance objectives that fundamentally conflict: shorter routes carry higher flight risk, low-risk corridors drain more battery, and energy-efficient paths are rarely the fastest. No single fixed trade-off serves all payload types, yet existing planners apply the same objective weights to every mission regardless of what is being carried.

Existing multi-objective planners—including entropy-based, diversity-driven, and crowding-distance Pareto selection methods—encode objective weights as a fixed scalar before path search begins. Payload properties such as fragility class, temperature sensitivity, and delivery urgency have no influence over which Pareto-optimal route is chosen. When the payload changes, operators must re-tune weights manually and re-run the planner—an intervention that is infeasible under field conditions.

This work proposes a *Payload-Aware Two-Tier UAV Route Planning* system that separates Pareto front construction from route selection, introducing an explicit algebraic bridge between Tier 1 routing objectives and Tier 2 payload metadata. Tier 1 runs a multi-objective A*/Dijkstra graph-search over a delivery airspace graph sourced from OpenStreetMap, producing a Pareto-optimal route set across distance, energy consumption, and flight risk. Tier 2 reads four operator-assigned payload tags and derives objective weights through a fixed formula: fragility class scales w_{risk}, delivery urgency scales w_{dist}, and temperature sensitivity scales w_{energy}. These weights score every Pareto point through a utility function; the $\arg\max$ is selected as the mission route.

The system will be evaluated on programmatically generated airspace graphs at three scales (50, 200, and 500 nodes) against single-objective A* and MOPSO baselines, using hypervolume, Pareto spread, and Tier 2 selection accuracy as metrics. Hardware validation will be conducted using a Pixhawk-equipped quadcopter through ArduPilot SITL simulation and outdoor flight tests.

The principal contribution is a formula-driven mapping from payload metadata to routing objective weights, connecting mission context directly to Pareto point selection without expert intervention. This replaces context-blind diversity criteria with a direct algebraic link from payload description to route choice, with the Pareto front computed once and reused across all mission types without re-running the planner.

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CHAPTER 1

INTRODUCTION AND PROBLEM STATEMENT

Unmanned aerial vehicles (UAVs) have emerged as a compelling platform for last-mile delivery in scenarios where ground transportation is impractical—medical supply chains to remote clinics, emergency drug drops during disaster response, and fragile diagnostic sample transport where road vibration risks contamination. In each scenario the UAV must fly a route from a dispatch hub to a delivery point across an airspace that contains obstacles, regulated no-fly zones, and variable wind corridors. A good route balances three conflicting physical objectives simultaneously: minimising total flight distance, minimising battery energy expenditure, and avoiding high-risk segments. These objectives are genuinely in tension—the shortest path crosses the most densely occupied airspace, the safest corridor requires significant detours, and the most energy-efficient altitude profile rarely coincides with either.

The standard engineering response to such a conflict is to form a weighted sum of the three objectives into a single scalar cost function, and then run a conventional path planner such as A* on it. This scalar collapses the multi-objective problem into a single-objective one, and the operator adjusts the three weights to express the mission's priority before each flight. The approach works for a fixed mission type, but it fails immediately when the payload changes: emergency medicine requires a completely different weight profile from fragile biological samples or disaster-relief bulk supplies. Manually re-tuning and re-running the planner between missions is operationally impractical in time-critical field conditions.

This chapter presents the formal problem statement, the research objectives the proposed system addresses, and the motivation for the two-tier architectural separation that is the core contribution of this work.

1.1 PROBLEM DOMAIN

Autonomous UAV systems for medical and emergency last-mile delivery operate under a structural constraint that distinguishes them from general routing problems: the optimal route trade-off is not fixed—it is a function of what is being carried. A fragile diagnostic sample demands a low-vibration, low-turbulence corridor even at the cost of a much longer flight time. Emergency insulin requires the shortest path regardless of risk. Bulk disaster-relief supplies demand the most energy-efficient route to maximise delivery range.

No current UAV route planner adjusts its routing priorities to match the payload being carried. The same fixed objective weights are applied to every mission, forcing field operators to manually recalibrate between every deployment. This is impractical and error-prone in time-critical emergency scenarios.

1.2 PROBLEM STATEMENT

Existing UAV route planners encode distance, energy consumption, and flight risk into a **single hand-tuned weighted-sum scalar** before path search begins, permanently fixing the objective trade-off regardless of what is being carried.

When the payload context changes—from fragile diagnostic samples requiring vibration-minimal routing to time-critical emergency medicine requiring shortest-path routing—an expert must manually re-tune the planner weights and re-run the search from scratch. This intervention is **infeasible in field conditions**.

The proposed system resolves this by separating Pareto front generation from mission-context-driven route selection. Payload metadata (fragility class, temperature sensitivity, delivery urgency) has never been used as the sole input to autonomous Pareto point selection in the UAV delivery literature, leaving a structural gap this project directly addresses.

1.3 MOTIVATION

The motivation for this work arises from three real operational scenarios that impose fundamentally incompatible routing requirements:

Fragile medical samples (tissue biopsies, blood samples, pathology specimens) are highly sensitive to vibration, g-forces, and thermal exposure. The dominant routing priority is minimising flight risk—defined here as exposure to turbulent corridors, sudden altitude changes, and wind-exposed segments. A route that adds 40% extra distance but halves the risk of vibration damage is operationally preferable.

Emergency medicine (insulin, epinephrine, antivenom) has a strict time window. The dominant routing priority is minimising route distance to minimise transit time. Flight risk and energy are secondary. For a payload with a 15-minute delivery window, the shortest path wins unconditionally.

Disaster-relief bulk supplies (water purification tablets, bandages, ration packs) weigh significantly more than medical payloads and are not time-critical or fragile. The dominant routing priority is energy efficiency—minimising battery drain to maximise the number of deliveries possible on a single charge cycle.

No existing UAV planner can switch between these three routing strategies based on a payload tag. The proposed two-tier system accomplishes this without any expert intervention or replanning.

1.4 RESEARCH OBJECTIVES

The following objectives guide this project:

1. Formulate multi-objective UAV route planning as a structured **Pareto front pre-computation** problem over a delivery airspace graph, theoretically distinguishing the deterministic graph-search approach from stochastic population-based alternatives.

2. Design a **payload metadata schema** and a principled rule-based mapping that translates mission context (fragility class, delivery urgency, temperature sensitivity) into Pareto point selection objective weights without any expert weight tuning.

3. **Benchmark Tier 1** route quality and Pareto front coverage against single-objective A* and MOPSO baselines on identical delivery scenarios, measuring hypervolume and Pareto spread (Inverted Generational Distance).

4. **Quantify Tier 2 selection accuracy** by measuring alignment between selected route characteristics and payload tag requirements across diverse mission types (fragile medical sample, emergency medicine, disaster-relief supplies).

5. Establish the system's **novelty**: payload metadata is used as the sole input to Pareto point selection, replacing context-blind entropy and diversity-based criteria with a deterministic algebraic mapping from payload description to route choice.

1.5 ORGANISATION OF THE REPORT

The remainder of this report is organised as follows. Chapter 2 presents a literature survey of three directly relevant multi-objective UAV path planning papers, identifies the common research gap they share, and states the research gaps this project addresses. Chapter 3 describes the proposed two-tier system architecture in detail, including pseudocode for both tiers, a architecture diagram of the full system, and separate module descriptions for Phase I (software) and Phase II (hardware). Chapter 4 explains the demonstration plan for both phases. Chapter 5 presents the project timeline as a Gantt chart. References are listed at the end.

CHAPTER 2

LITERATURE SURVEY

2.1 OVERVIEW

Multi-objective UAV path planning has attracted sustained research attention because no single route can simultaneously minimise distance, energy, and flight risk—these objectives are structurally in conflict. The dominant approach in the literature is to generate a **Pareto-optimal front** of non-dominated routes, each representing a different trade-off across the objective space, and then select one route from that front as the mission route.

The critical question that determines the practical value of any such system is: *how is the final route selected from the Pareto front?* The three most directly relevant papers—reviewed in detail below—all answer this question using **front diversity metrics**: entropy, geometric spread, or crowding distance. These metrics select routes that are well-distributed across the Pareto front, but they are entirely blind to the delivery context. None of the three papers allow what the UAV is carrying to influence which route is chosen.

This is the structural research gap this project addresses.

2.2 REVIEW OF RELATED WORK

2.2.1 Chokkalingam and Ganesan (2026)

Chokkalingam and Ganesan [1] present a multi-objective UAV path planning framework that uses adaptive constraint handling combined with an **entropy-based criterion** to select a final route from the generated Pareto front. The entropy metric measures the information content of the objective distribution across Pareto points: a point with high entropy is selected because it maximises the diversity of objective trade-offs available to the operator.

The system is evaluated on custom simulation environments and demonstrates strong Pareto front coverage compared to standard MOPSO baselines. The entropy criterion is shown to select routes that span a wider range of objective combinations than random or distance-ranked selection.

Limitation. The entropy criterion selects solutions based on front diversity alone. Payload metadata—fragility class, delivery urgency, thermal sensitivity—is never incorporated into the selection process. An operator carrying fragile biological samples and an operator carrying emergency insulin would receive the same selected route from the same Pareto front, because the entropy of the front is independent of what is being transported.

2.2.2 Datsko et al. (2024)

Datsko et al. [2] propose a diversity-driven Pareto selection mechanism for UAV routing that optimises **geometric spread** of candidate solutions across the

multi-objective space. The key idea is to select Pareto points that maximise the physical volume of the objective space that is "covered" by the candidate solution set, ensuring the operator has access to a geometrically diverse set of route options.

The paper demonstrates improvements in Inverted Generational Distance (IGD) over diversity-agnostic selection methods and shows that geometrically spread solutions provide better coverage of the true Pareto front when the true front is not known in advance.

Limitation. The geometric-spread selection mechanism is guided entirely by solution diversity in objective space. The delivery context—payload type, time window, priority class—has no influence on which route is chosen. The mechanism is designed to serve human decision-making (giving the operator a diverse set of choices) rather than autonomous selection aligned with mission requirements.

2.2.3 Ntakolia et al. (2023)

Ntakolia et al. [3] apply a **crowding-distance-based** Pareto point selection method within a multi-objective UAV path planner. Crowding distance, originally introduced in NSGA-II, measures how isolated each Pareto point is from its neighbours in objective space: points with high crowding distance are in less dense regions of the front and are preferred to maintain spread.

The paper shows that crowding-distance selection produces well-spread Pareto fronts for UAV routing problems and compares favourably against NSGA-II and ε-constraint methods on grid-world airspace environments.

Limitation. Crowding distance is a purely geometric measure of front density. It cannot encode mission-specific requirements. Payload fragility, temperature sensitivity, or urgency class cannot influence route selection at any stage. Like entropy and geometric-spread methods, crowding distance maximises front diversity without any awareness of what is actually being delivered.

2.3 LITERATURE SURVEY SUMMARY TABLE

Table 2.1 summarises all three papers across key dimensions relevant to the proposed work.

2.4 RESEARCH GAPS

2.4.1 Common Gap Across All Three Papers

All three reviewed papers select Pareto-optimal routes based on **front diversity metrics**—entropy, geometric spread, and crowding distance respectively. None of them allow the delivery context (what payload is being carried) to influence which route is selected. This is the structural gap the proposed system directly addresses.

The gap is not a minor implementation detail—it is a fundamental design assumption. Entropy, geometric spread, and crowding distance are all defined

TABLE 2.1: Summary of Reviewed Multi-Objective UAV Path Planning Literature

Author(s) & Year	**Journal / Venue**	**Technique**	**Pareto Selection Method**	**Limitation / Research Gap**
Chokkalingam & Ganesan (2026) [1]	RAIRO – Oper. Res.	Multi-objective UAV path planning with adaptive constraint handling	Entropy-based criterion (front diversity)	Payload metadata never incorporated; fragile samples and emergency medicine receive identical route selection
Datsko et al. (2024) [2]	IEEE RA-L	Diversity-driven Pareto selection for UAV routing	Geometric spread (maximise volume of objective space covered)	Selection is guided entirely by solution diversity; delivery context has no influence on route choice
Ntakolia et al. (2023) [3]	J. Intell. Robot. Syst.	Crowding-distance Pareto selection in multi-objective UAV planner	Crowding distance (NSGA-II style)	Crowding distance cannot encode urgency, fragility, or thermal sensitivity; mission context is absent from selection

over the Pareto front itself. They have no interface through which external mission context can enter. Extending these methods to incorporate payload information would require replacing the selection criterion entirely, which is precisely what the proposed Tier 2 utility scoring mechanism does.

2.4.2 Specific Gaps Addressed by This Work

Gap 1 — No payload-aware Pareto selection. All existing UAV multi-objective planners apply a fixed selection criterion independent of payload type. The proposed Tier 2 introduces a formula-driven mapping from four payload tags directly to Pareto point selection.

Gap 2 — No re-use across mission types. Existing planners re-run the full multi-objective search when the mission context changes. The proposed two-tier separation pre-computes the Pareto front once (Tier 1) and allows instantaneous route selection for any new payload tag (Tier 2), without re-running the planner.

Gap 3 — No algebraic bridge between delivery context and routing objectives. The mapping between a payload description and the routing trade-off it implies (e.g., fragility $\rightarrow$ minimise risk, urgency $\rightarrow$ minimise distance) has never been formalised algebraically. The proposed weight-derivation formula provides this bridge, making the mapping deterministic, transparent, and reproducible.

CHAPTER 3

PROPOSED METHODOLOGY

3.1 ARCHITECTURAL OVERVIEW

The proposed system is built on a single architectural decision: **separate Pareto front construction (Tier 1) from route selection (Tier 2)**. This separation eliminates the need to re-run a multi-objective search each time the payload changes. The Pareto front is computed once for a delivery region and stored. For each new mission, only Tier 2 runs—reading the payload tag and selecting the appropriate route from the pre-computed front in near-instantaneous time.

Figure 3.1 shows the complete system architecture.

3.2 TIER 1 — DETERMINISTIC PARETO FRONT CONSTRUCTION

3.2.1 What Tier 1 Does

Tier 1 implements a multi-objective A*/Dijkstra graph-search over a delivery airspace graph. The output is a **set of Pareto-optimal routes** where no single route dominates another on all three objectives simultaneously. This front is computed once for a given delivery region and stored persistently; it does not need to be recomputed between missions.

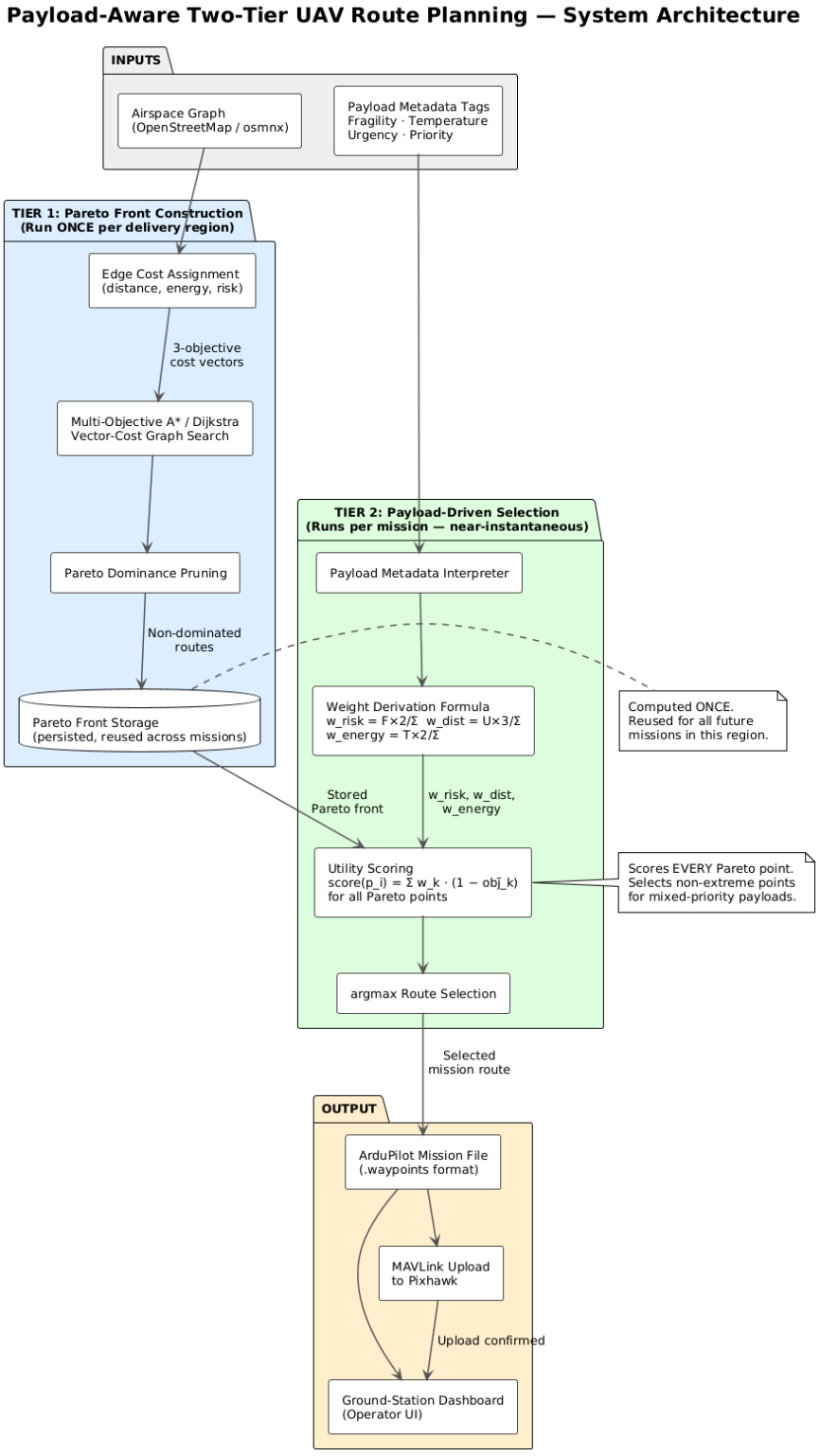


FIGURE 3.1: Overall System Architecture: Two-Tier Payload-Aware UAV Route Planner. Tier 1 pre-computes the Pareto front once; Tier 2 selects the optimal route for each new payload tag using the stored front.

3.2.2 Airspace Graph Construction

The airspace graph $G = (V, E)$ is extracted from OpenStreetMap using the `osmnx` Python library [4]. Each edge $e \in E$ represents a flight segment between two waypoints and carries a **cost vector**:

$$\mathbf{c}(e) = \left[d(e),\ \varepsilon(e),\ \rho(e) \right] \tag{3.1}$$

where:

- $d(e)$ = geodesic distance of the flight segment (km)
- $\varepsilon(e)$ = estimated energy consumption (proxy: $d(e) \times$ altitude-change factor)
- $\rho(e)$ = flight risk score $\in [0, 1]$, derived from obstacle density, proximity to no-fly zones, and wind exposure (simulated programmatically for reproducible benchmarking)

3.2.3 Three Optimisation Objectives

The three routing objectives are:

1. **Minimise route distance** d: total geodesic path length in km.
2. **Minimise energy consumption** ε: total estimated battery usage as a percentage of full charge.

3. **Minimise flight risk** ρ: average risk score across all flight segments, where risk captures obstacle density, regulated airspace proximity, and wind exposure.

A route p dominates route q if p is at least as good as q on all three objectives and strictly better on at least one. The Pareto front $\mathcal{P}$ is the set of all non-dominated routes.

3.2.4 Tier 1 Activity Flow

Figure 3.2 shows the step-by-step flow of Tier 1.

3.2.5 Pseudocode: Multi-Objective A* for Pareto Front Construction

Algorithm 1 presents the pseudocode for Tier 1.

Key properties. The algorithm maintains a cost vector $\mathbf{g}$ (not a scalar) for each partial path and prunes paths that are dominated by already-discovered routes. `UpdateParetoFront` adds the new route and removes any previously stored routes it dominates. The result is a deterministic, reproducible Pareto front that can be stored and re-used across all missions in a deployment region.

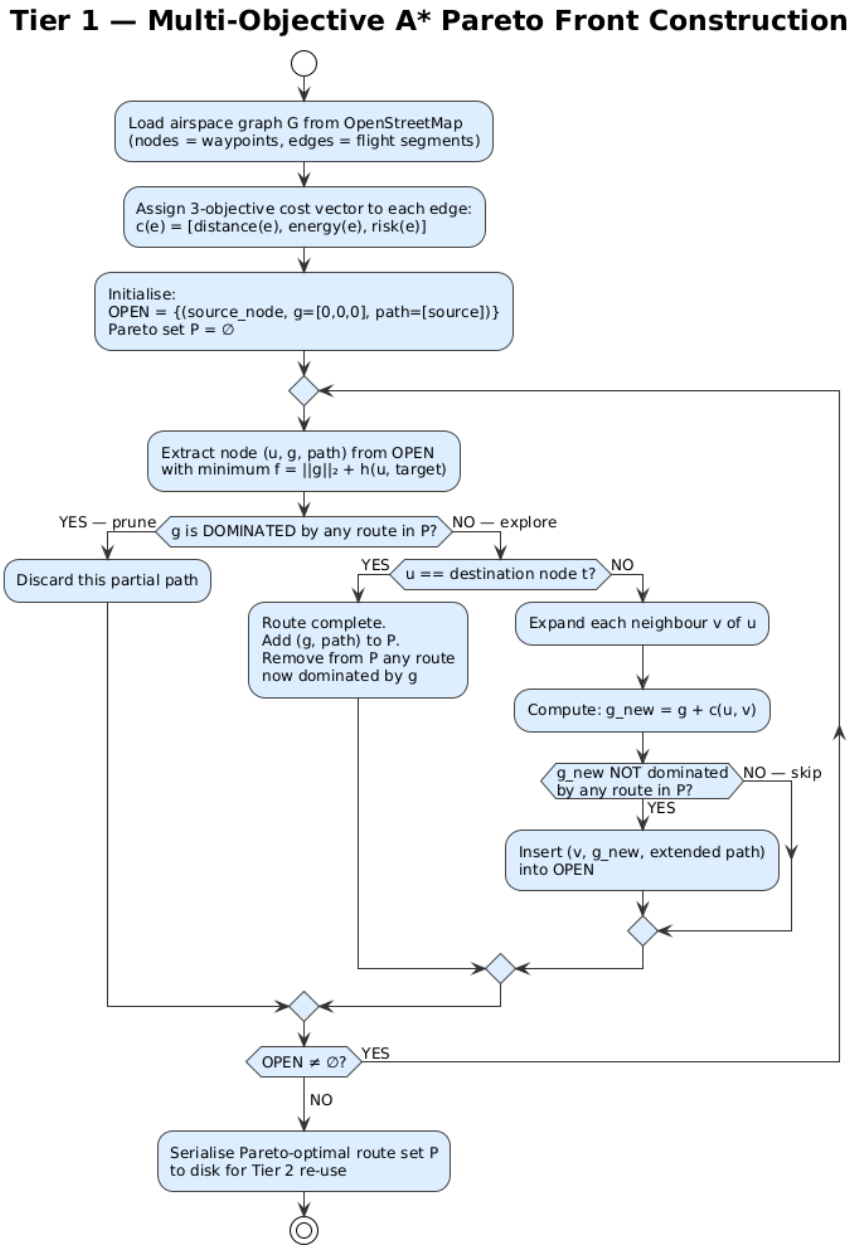


FIGURE 3.2: Tier 1 Activity Flow: Multi-Objective A* Pareto Front Construction. The open set is explored in order of estimated total cost until no further non-dominated routes exist.

Algorithm 1 Multi-Objective A* Pareto Front Construction (Tier 1)

Input: Airspace graph $G = (V, E)$ with edge cost vectors $\mathbf{c}(e) = [d(e), \varepsilon(e), \rho(e)]$, source node s, destination node t
Output: Pareto-optimal route set $\mathcal{P}$

OPEN $\leftarrow \{ (s,\ \mathbf{g} = [0,0,0],\ \text{path} = [s]) \}$
$\mathcal{P} \leftarrow \varnothing$ ▷ Pareto front (empty initially)
while OPEN $\neq \varnothing$ **do**
    $(u, \mathbf{g}, \text{path}) \leftarrow \text{ExtractMin}(\text{OPEN})$ ▷ Min by $f = \|\mathbf{g}\|_2 + h(u,t)$
    if $\mathbf{g}$ is dominated by any route in $\mathcal{P}$ **then**
        continue ▷ Prune dominated paths early
    end if
    if $u = t$ **then**
        $\mathcal{P} \leftarrow \text{UpdateParetoFront}(\mathcal{P}, \mathbf{g}, \text{path})$ ▷ Remove newly dominated routes from $\mathcal{P}$
    else
        for each neighbour v of u in G **do**
            $\mathbf{g}_{\text{new}} \leftarrow \mathbf{g} + \mathbf{c}(u,v)$
            if $\mathbf{g}_{\text{new}}$ is **not** dominated by any route in $\mathcal{P}$ **then**
                $\text{OPEN.insert}\big(v,\ \mathbf{g}_{\text{new}},\ \text{path} \cup \{v\}\big)$
            end if
        end for
    end if
end while
return $\mathcal{P}$ ▷ Stored to disk for Tier 2 re-use

3.3 THE TIER 1–TIER 2 CONNECTION: THE ALGEBRAIC BRIDGE

The connection between Tier 1 and Tier 2 is the central design decision of this project. This section makes it explicit.

3.3.1 Semantic Mapping: From Payload Tags to Routing Objectives

Each of the four payload metadata tags maps to exactly one Tier 1 routing objective:

TABLE 3.1: Semantic Mapping: Payload Metadata Tags to Tier 1 Routing Objectives

Payload Tag	**Range**	**Maps to Tier 1 Objective**	**Reasoning**
Fragility class F	1–5	w_{risk} (flight risk weight)	Fragile items need low-vibration, low-risk corridors
Temperature sensitivity T	1–3	w_{energy} (energy weight)	Temp-sensitive items need short exposure time $\rightarrow$ fast flight
Delivery urgency U	1–3	w_{dist} (distance weight)	Time-critical deliveries need the shortest path
Priority level P	1–5	Tie-breaker amplifier	Scales the dominant weight when scores are close

3.3.2 Weight Derivation Formula

From the payload tags, the three routing objective weights are derived **algorithmically** with no expert input:

$$r_{\text{raw}} = F \times 2 \tag{3.2}$$

$$d_{\text{raw}} = U \times 3 \tag{3.3}$$

$$e_{\text{raw}} = T \times 2 \tag{3.4}$$

$$w_{\text{risk}} = \frac{r_{\text{raw}}}{r_{\text{raw}} + d_{\text{raw}} + e_{\text{raw}}} \tag{3.5}$$

$$w_{\text{dist}} = \frac{d_{\text{raw}}}{r_{\text{raw}} + d_{\text{raw}} + e_{\text{raw}}} \tag{3.6}$$

$$w_{\text{energy}} = \frac{e_{\text{raw}}}{r_{\text{raw}} + d_{\text{raw}} + e_{\text{raw}}} \tag{3.7}$$

The multipliers (2 for fragility and temperature, 3 for urgency) reflect the domain insight that urgency has a higher penalty sensitivity than fragility or thermal constraints in the UAV medical delivery context.

3.3.3 Why This Is Not Extreme-Point Selection

A critical property of the utility scoring approach (Section 3.4) is that it selects **different, non-extreme points** for most real payloads. Table 3.2 demonstrates this for four representative payload presets.

Only emergency medicine selects an extreme point—and that is the *correct* answer for that payload. All other mixed-priority payloads select intermediate, non-extreme points that neither single-objective A* nor MOPSO can produce.

TABLE 3.2: Payload Presets and Expected Pareto Point Selection

Payload Preset	F	T	U	w_{risk}	w_{dist}	w_{energy}	**Expected Selected Point**
Fragile medical sample	5	1	1	0.667	0.200	0.133	Non-extreme: long, low-risk (NOT the extreme lowest-risk)
Emergency medicine	1	1	3	0.154	0.692	0.154	Extreme shortest-distance (correct for this payload)
Disaster-relief supplies	1	2	1	0.222	0.333	0.444	Non-extreme: energy-optimal intermediate point
Mixed cargo	3	2	2	0.375	0.375	0.250	Non-extreme: balanced trade-off

3.4 TIER 2 — PAYLOAD-DRIVEN PARETO SELECTION

3.4.1 What Tier 2 Does

Tier 2 reads the four payload metadata tags assigned by the operator, derives the three routing objective weights via the formula in Section 3.3, normalises the Pareto front objectives, computes a utility score for every Pareto point, and returns the arg max as the mission route. No weight re-tuning occurs; no planner is re-run.

3.4.2 Tier 2 Activity Flow

Figure 3.3 shows the step-by-step flow of Tier 2.

3.4.3 Utility Scoring Formula

For each Pareto point p_i with objective values $(d_i, \varepsilon_i, \rho_i)$, the normalised objective values are:

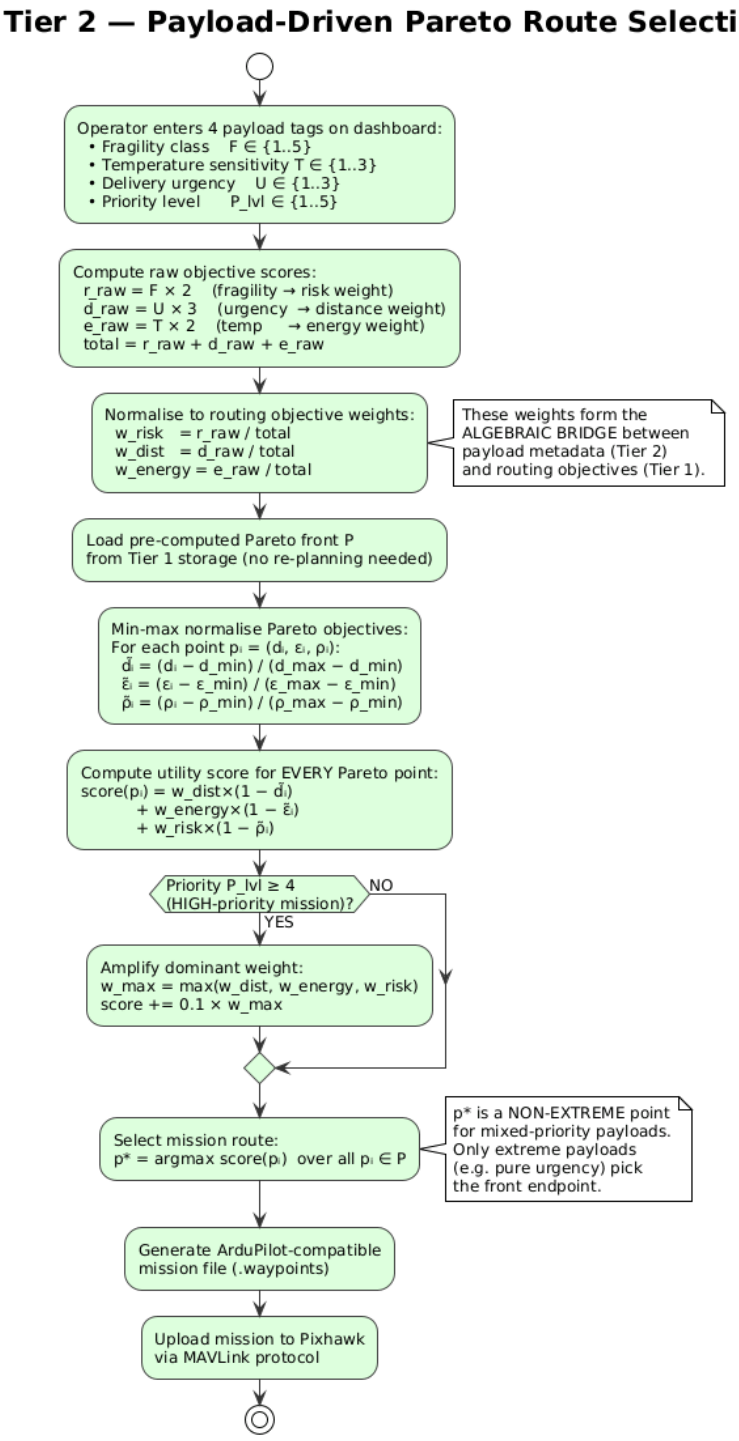


FIGURE 3.3: Tier 2 Activity Flow: Payload-Driven Pareto Selection. Payload tags are algebraically mapped to weights, which score every Pareto point via a utility function; the argmax is selected as the mission route.

$$\tilde{d}_i = \frac{d_i - d_{\min}}{d_{\max} - d_{\min}}, \quad \tilde{\varepsilon}_i = \frac{\varepsilon_i - \varepsilon_{\min}}{\varepsilon_{\max} - \varepsilon_{\min}}, \quad \tilde{\rho}_i = \frac{\rho_i - \rho_{\min}}{\rho_{\max} - \rho_{\min}} \tag{3.8}$$

where $d_{\min}, d_{\max}$, etc., are the extremes over the full Pareto front $\mathcal{P}$. The utility score is:

$$\text{score}(p_i) = w_{\text{dist}} \cdot (1 - \tilde{d}_i) + w_{\text{energy}} \cdot (1 - \tilde{\varepsilon}_i) + w_{\text{risk}} \cdot (1 - \tilde{\rho}_i) \tag{3.9}$$

Selected route:

$$p^* = \arg\max_{p_i \in \mathcal{P}} \text{score}(p_i) \tag{3.10}$$

A high w_{risk} (from a high fragility tag) drives the score to favour points with low $\tilde{\rho}_i$ (low risk). A high w_{dist} (from a high urgency tag) drives the score to favour points with low $\tilde{d}_i$ (short distance). The formula scores the *entire* Pareto front, not just the extremes.

3.4.4 Pseudocode: Payload-Driven Route Selection

Algorithm 2 presents the complete Tier 2 pseudocode.

3.5 MODULES FOR PHASE I (SOFTWARE SIMULATION)

Phase I delivers a complete, validated software system. The following modules are implemented, tested, and benchmarked entirely in simulation:

Module 1 — Airspace Graph Builder. Extracts the road/path network from OpenStreetMap for a specified delivery region using `osmnx`. Assigns three

Algorithm 2 Payload-Driven Pareto Route Selection (Tier 2)

Input: Payload tags: Fragility $F \in \{1..5\}$, Temperature $T \in \{1..3\}$, Urgency $U \in \{1..3\}$, Priority $P_{\text{lvl}} \in \{1..5\}$
Input: Pre-computed Pareto front $\mathcal{P} = \{(d_i, \varepsilon_i, \rho_i)\}_{i=1}^{k}$
Output: Selected mission route p^*

— Step 1: Derive raw scores from payload tags —
$r_{\text{raw}} \leftarrow F \times 2$
$d_{\text{raw}} \leftarrow U \times 3$
$e_{\text{raw}} \leftarrow T \times 2$
$\text{total} \leftarrow r_{\text{raw}} + d_{\text{raw}} + e_{\text{raw}}$
— Step 2: Normalise to objective weights —
$w_{\text{risk}} \leftarrow r_{\text{raw}}/\text{total}$
$w_{\text{dist}} \leftarrow d_{\text{raw}}/\text{total}$
$w_{\text{energy}} \leftarrow e_{\text{raw}}/\text{total}$
— Step 3: Min-max normalise Pareto objectives —
$(d_{\min}, d_{\max}) \leftarrow (\min_{p_i \in \mathcal{P}} d_i,\ \max_{p_i \in \mathcal{P}} d_i)$
$(\varepsilon_{\min}, \varepsilon_{\max}) \leftarrow (\min_{p_i} \varepsilon_i,\ \max_{p_i} \varepsilon_i)$
$(\rho_{\min}, \rho_{\max}) \leftarrow (\min_{p_i} \rho_i,\ \max_{p_i} \rho_i)$
— Step 4: Compute utility score for each Pareto point —
$\text{best_score} \leftarrow -\infty; \quad p^* \leftarrow \text{null}$
for each $p_i = (d_i, \varepsilon_i, \rho_i) \in \mathcal{P}$ **do**
  $\tilde{d}_i \leftarrow (d_i - d_{\min})/(d_{\max} - d_{\min})$
  $\tilde{\varepsilon}_i \leftarrow (\varepsilon_i - \varepsilon_{\min})/(\varepsilon_{\max} - \varepsilon_{\min})$
  $\tilde{\rho}_i \leftarrow (\rho_i - \rho_{\min})/(\rho_{\max} - \rho_{\min})$
  $\text{score}_i \leftarrow w_{\text{dist}} \cdot (1 - \tilde{d}_i) + w_{\text{energy}} \cdot (1 - \tilde{\varepsilon}_i) + w_{\text{risk}} \cdot (1 - \tilde{\rho}_i)$
  if $P_{\text{lvl}} \geq 4$ **then** ▷ High-priority: amplify dominant weight
    $w_{\max} \leftarrow \max(w_{\text{dist}}, w_{\text{energy}}, w_{\text{risk}})$
    $\text{score}_i \leftarrow \text{score}_i + 0.1 \times w_{\max}$
  end if
  if $\text{score}_i > \text{best_score}$ **then**
    $\text{best_score} \leftarrow \text{score}_i; \quad p^* \leftarrow p_i$
  end if
end for
return p^*

objective costs to each edge: distance (geodesic), energy (distance $\times$ altitude-change factor), and flight risk (simulated threat zones and obstacle

density scored 0–1 using a configurable hazard map).

Module 2 — Multi-Objective A*/Dijkstra Engine (Tier 1). Implements Algorithm 1 using Python and NetworkX [5]. Produces a Pareto-optimal route set stored as a serialised Python object. The module includes unit tests that verify Pareto non-dominance of all returned routes.

Module 3 — MOPSO Baseline. Implements a Multi-Objective Particle Swarm Optimisation baseline [6] with swarm size 100, 200 iterations, and a Pareto archive of 100 solutions. Used to benchmark Tier 1 on hypervolume and Pareto spread.

Module 4 — Payload Metadata Interpreter (Tier 2). Implements Algorithm 2: reads the four payload tags, derives objective weights via Equations (3.5)–(3.7), applies utility scoring (Equation (3.9)), and returns the arg max route.

Module 5 — Benchmarking Suite. Evaluates all methods on three scenario scales: S1 ($\sim$50 nodes, low obstacle density), S2 ($\sim$200 nodes, medium density), and S3 ($\sim$500 nodes, high density). Metrics: Hypervolume (HV), Inverted Generational Distance (IGD), and Tier 2 selection accuracy across four payload presets. MOPSO runs 30 times per scenario to obtain mean $\pm$ standard deviation.

Module 6 — Ground-Station Dashboard. A Tkinter or React operator interface where the operator enters the four payload tags once. The system automatically selects the route from the pre-computed Pareto front and generates an ArduPilot-compatible mission file (`.waypoints`) for upload to the UAV.

3.6 MODULES FOR PHASE II (HARDWARE IMPLEMENTATION)

Phase II deploys the validated software system onto physical hardware and conducts outdoor flight tests:

Module 7 — Raspberry Pi 5 Integration. Ports the Tier 1 and Tier 2 Python modules to the onboard Raspberry Pi 5. Verifies that Tier 1 pre-computation runs within acceptable time on the onboard CPU, and that Tier 2 route selection is near-instantaneous ($<$1 s end-to-end from payload tag input to mission file output).

Module 8 — MAVLink Mission Upload. Implements MAVLink [7] communication between the Raspberry Pi and the Pixhawk flight controller. Converts the selected Pareto route into ArduPilot-compatible waypoint format and uploads it to the Pixhawk via the MAVLink protocol in real time.

Module 9 — ArduPilot SITL Validation (Pre-flight). Before any real flight, all routes selected by Tier 2 are validated in ArduPilot's Software-in-the-Loop (SITL) simulation. SITL faithfully emulates the Pixhawk's flight controller software, allowing end-to-end validation of the route adherence behaviour without hardware risk.

Module 10 — Outdoor Flight Tests. Real outdoor flights on the F450 quadcopter frame with Pixhawk flight controller. Measures route adherence (GPS position error vs. planned route), end-to-end latency (payload tag entry to mission upload), and payload delivery success rate across all three payload presets (fragile, emergency, disaster-relief).

3.7 TOOLS AND TECHNOLOGY STACK

Table 3.3 summarises the full tools and software stack.

TABLE 3.3: Tools and Technology Stack

Component	**Tool / Library**
Graph construction	osmnx (Python), OpenStreetMap data
Multi-objective search	Python, NetworkX [5]
Objective computation	NumPy
Pareto front storage	Python serialisation (pickle / JSON)
Visualisation	Matplotlib, custom interactive HTML
MOPSO baseline	Python (custom implementation) [6]
UAV mission upload	ArduPilot Mission Planner, MAVLink [7, 8]
Ground-station UI	Tkinter (Phase I); React (optional Phase II)
Onboard compute	Raspberry Pi 5
Flight controller	Pixhawk (ArduPilot firmware)
UAV frame	F450 quadcopter

CHAPTER 4

DEMONSTRATION PLAN

This chapter describes how the proposed system will be demonstrated to the project panel. The demonstration follows a two-phase structure: Phase I demonstrates the complete algorithmic pipeline through software simulations, and Phase II extends the validated software to hardware deployment and outdoor flight tests.

4.1 PHASE I DEMONSTRATION — SOFTWARE SIMULATION

Phase I demonstrations are entirely software-based. All demos run on a standard laptop with no GPU or cloud dependency. The goal of Phase I demos is to let the panel **see the algorithm making decisions in real time** and to compare the proposed system against baselines with quantitative output.

4.1.1 Demo 1: Airspace Graph and Objective Visualisation

What is shown. A live visualisation of the extracted OpenStreetMap airspace graph for a chosen delivery region. Edge colours represent the three objective costs: blue for distance, green for energy, red for risk. The operator can toggle between objective overlays and zoom into specific delivery segments.

Purpose. Establishes that the Tier 1 inputs (the three-cost-vector graph) are correctly constructed and visually interpretable. The panel can see how obstacle zones, no-fly corridors, and wind-exposed segments translate into high-risk edge colours.

Tools. Python (`osmnx`, `networkx`, `matplotlib`).

4.1.2 Demo 2: Tier 1 Pareto Front Generation and 3D Scatter Plot

What is shown. The multi-objective A* algorithm is run on the medium-scale scenario (S2, $\sim$200 nodes). The resulting Pareto front is visualised as an interactive 3D scatter plot with axes: route distance, energy consumption, and flight risk. All non-dominated routes are shown as points. The MOPSO baseline front is overlaid in a contrasting colour for visual comparison.

Purpose. Demonstrates that Tier 1 produces a well-distributed Pareto front with higher hypervolume (HV) than MOPSO. The panel can visually verify that no point in the proposed system's front is dominated by any point in the MOPSO front.

Tools. Python (`matplotlib` 3D scatter, or `plotly` for interactive rotation).

4.1.3 Demo 3: Tier 2 Payload-Aware Route Selection (Interactive)

What is shown. The ground-station dashboard interface is opened. The operator enters payload tags for three scenarios in sequence:

1. **Fragile medical sample**: $F = 5, T = 1, U = 1, P = 3$
2. **Emergency medicine**: $F = 1, T = 1, U = 3, P = 5$
3. **Disaster-relief supplies**: $F = 1, T = 2, U = 1, P = 3$

For each tag entry, the system instantly:

- Displays the derived weights $(w_{\text{risk}}, w_{\text{dist}}, w_{\text{energy}})$
- Highlights the selected Pareto point on the 3D scatter plot
- Overlays the selected route on the airspace graph map
- Shows the mission statistics: distance, estimated energy use, and risk score

Purpose. Provides the most direct visual demonstration of the core contribution. The panel can see the selected point **move to a different, non-extreme location** on the Pareto front for each payload type, confirming that the utility scoring selects genuinely different trade-offs.

Tools. Tkinter or React dashboard; `matplotlib` or `plotly` for real-time scatter highlight.

4.1.4 Demo 4: Quantitative Benchmark Comparison

What is shown. A table and bar chart comparing the proposed Tier 1 system against single-objective A* and MOPSO on all three scenario scales (S1, S2, S3). Metrics displayed: Hypervolume (HV), Inverted Generational Distance (IGD), and computation time. MOPSO results include mean $\pm$ standard deviation over 30 runs.

Purpose. Provides the panel with rigorous quantitative evidence that Tier 1 produces a higher-quality Pareto front than both baselines, particularly on hypervolume (which measures the volume of dominated objective space—a higher value is strictly better).

Tools. Python benchmarking suite; `matplotlib` bar charts.

4.1.5 Demo 5: Tier 2 Selection Accuracy Measurement

What is shown. For each of the four payload presets (Table 3.2), the system's selected route is compared against the *ground-truth optimal* route (determined by exhaustive evaluation of all Pareto points with oracle weights). The **Tier 2 selection accuracy** is reported as the percentage of payload types for which the algorithm selected the same route as the oracle.

Purpose. Quantifies the alignment between payload-derived weights and the routing requirements they express, establishing that the algebraic mapping is not just principled in theory but correct in practice.

Tools. Python evaluation script; tabular output.

4.2 PHASE II DEMONSTRATION — HARDWARE IMPLEMENTATION

Phase II demonstrations are conducted once the software system has been fully validated in Phase I. All Phase I algorithmic validation is completed before any hardware is powered on. Real flight tests confirm an already-validated system, not the primary evaluation environment.

4.2.1 Step 1: ArduPilot SITL Validation

Before outdoor flight, every payload preset is tested in ArduPilot's Software-in-the-Loop (SITL) simulation. SITL runs the exact Pixhawk firmware on a desktop PC and emulates the physical flight controller. The selected Pareto route is uploaded as a waypoint file; SITL simulates the full flight and reports GPS track, battery consumption profile, and route adherence.

Pass criterion. Route adherence within $\pm 2\,\text{m}$ of planned waypoints; mission completion rate 100%; mission upload time $<30\,\text{s}$ end-to-end.

4.2.2 Step 2: Hardware Integration on Raspberry Pi 5

The complete Python pipeline (graph loader, Tier 1 pre-computation, Tier 2 selector, MAVLink uploader) is deployed to the Raspberry Pi 5 mounted on the F450 quadcopter frame. Integration tests verify that:

- Tier 2 route selection and mission file generation complete in $<1\,\text{s}$.

- MAVLink mission upload to Pixhawk completes reliably.
- The Pixhawk acknowledges and stores the waypoint mission correctly.

4.2.3 Step 3: Outdoor Flight Tests

Real flights are conducted on the F450 quadcopter with a servo-actuated payload release mechanism. Three flights are planned, one per payload preset. Each flight:

1. Operator enters payload tags on the ground-station UI.
2. Tier 2 selects route; mission file generated and uploaded via MAVLink.
3. UAV executes the mission autonomously.
4. GPS track is compared against the planned route.
5. Payload is released at the delivery point.

Measured metrics. End-to-end latency (tag entry to mission upload), route adherence (average cross-track error), and payload delivery success.

4.2.4 Summary

Table 4.1 summarises the demonstration structure across both phases.

TABLE 4.1: Demonstration Plan Summary

Phase	**Demo**	**What is Shown**	**Environment**
Phase I	1	Airspace graph & objective visualisation	Simulation
Phase I	2	3D Pareto front vs MOPSO baseline	Simulation
Phase I	3	Interactive payload-to-route demo	Dashboard / Laptop
Phase I	4	Quantitative benchmark table	Simulation
Phase I	5	Tier 2 selection accuracy measurement	Simulation
Phase II	6	ArduPilot SITL end-to-end validation	SITL / Desktop
Phase II	7	Raspberry Pi hardware integration test	Onboard compute
Phase II	8	Outdoor flight tests (3 payload types)	Real UAV outdoor

CHAPTER 5

PROJECT TIMELINE

The project spans four quarters from July 2026 to March 2027, aligned with the academic calendar of Phase I and Phase II. Figure 5.1 presents the Gantt chart for the full project timeline.

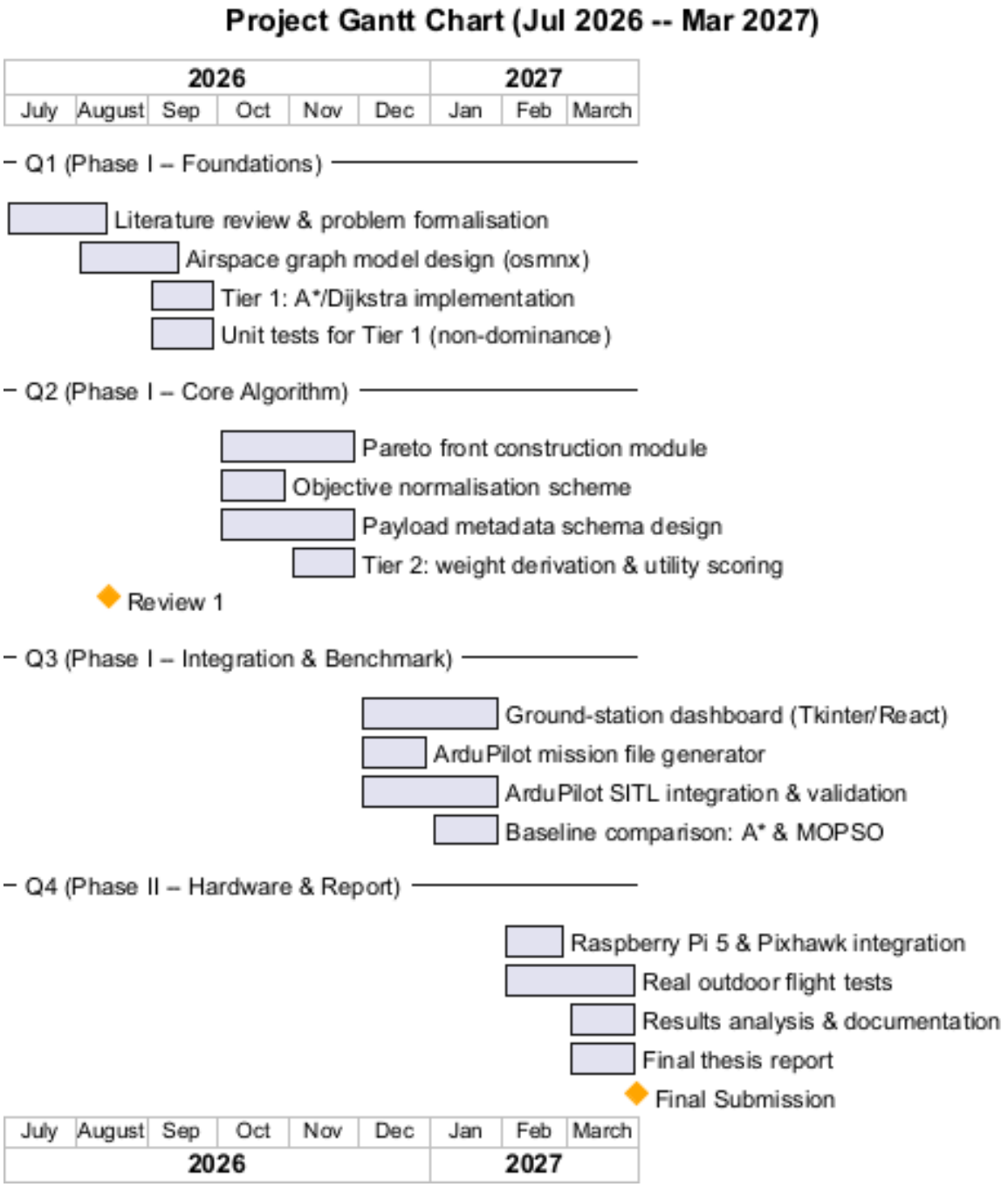


FIGURE 5.1: Project Gantt Chart (July 2026 – March 2027), generated with PlantUML (see `diagrams/gantt.puml`). Orange diamonds mark review milestones.

REFERENCES

1. M. Chokkalingam and K. Ganesan. Multi-objective UAV path planning using adaptive constraint handling with entropy-based criteria for Pareto front solution selection. *RAIRO – Operations Research*, 2026. doi: 10.1051/ro/2025159.

2. O. Datsko, M. Kholiavchenko, O. Ivanov, and S. Kholiavchenko. Diversity-driven Pareto selection for multi-objective UAV route planning. *IEEE Robotics and Automation Letters*, 2024. doi: 10.1109/LRA.2024.10414185. Verify full author list and exact title on IEEE Xplore.

3. C. Ntakolia, A. Deva, and F. Asaro. A crowding-distance-based multi-objective evolutionary algorithm for UAV path planning under multiple constraints. *Journal of Intelligent & Robotic Systems*, 107, 2023. doi: 10.1007/s10846-023-01945-9. Verify full author list and exact title via the DOI link.

4. Geoff Boeing. OSMnx: New methods for acquiring, constructing, analyzing, and visualizing complex street networks. *Computers, Environment and Urban Systems*, 65:126–139, 2017. doi: 10.1016/j.compenvurbsys.2017.05.004.

5. Aric A. Hagberg, Daniel A. Schult, and Pieter J. Swart. Exploring network structure, dynamics, and function using NetworkX. In *Proceedings of the 7th Python in Science Conference (SciPy2008)*, pages 11–15, Pasadena, CA, USA, 2008. URL `https://networkx.org`.

6. Carlos A. Coello Coello, Gregorio Toscano Pulido, and Maximino Salazar Lechuga. Handling multiple objectives with particle swarm optimization. *IEEE Transactions on Evolutionary Computation*, 8(3):256–279, 2004. doi: 10.1109/TEVC.2004.826067.

7. Lorenz Meier, Julian Camacho, Ben Godbolt, Kevin Usher, James Goppert, Pablo Castillo, and John Leffly. MAVLink: Micro air vehicle communication protocol. https://mavlink.io/en/, 2026. Accessed: August 2026.

8. ArduPilot Development Team. ArduPilot – open source autopilot. https://ardupilot.org, 2026. Accessed: August 2026.